

Beyond Statistical Relevance: A Domain-Aware Framework for Clinically Guided Feature Selection

Solving the Trustworthiness Gap via a Decoupled Operator-Driven Architecture (DODA)

Doctoral / Clinical AI Research Proposal

Focus Area: Clinical Machine Learning, Model Trustworthiness, Interpretability **Core Architecture:** Decoupled Operator-Driven Architecture (DODA)

1. Introduction & Problem Statement

Modern clinical machine learning models heavily rely on feature selection to reduce dimensionality, mitigate the risk of overfitting, and enhance computational efficiency. Standard pipelines rely on conventional statistical metrics — such as ANOVA, LASSO, Mutual Information, mRMR, Random Forest Feature Importance, and Boruta — to isolate optimal subsets. These methods evaluate variables strictly by optimizing mathematical alignments and mathematical associations with the target outcome.

While this convention improves standard out-of-sample predictive performance metrics, it presents a critical flaw in healthcare settings: statistical relevance does not inherently imply clinical meaningfulness. Consequently, pure algorithmic filters generate severe bottlenecks when translated to actual clinical deployments:

- **Discarding Crucial Biomarkers:** True biological or clinical signals that have slight statistical variances or are nonlinear across sub-cohorts are frequently pruned away.
- **Preference for False Dominance:** Statistically dominant, highly correlated confounding variables are favored over weak but causative biological indicators.
- **The Trust Deficit:** Resulting models lack clinical face validity, causing healthcare providers and clinical practitioners to reject their outputs.

Ultimately, this divergence restricts the clinical safety, regulatory approval, and practical utility of machine learning models in modern medicine.

2. Research Gap & Defensible Position

Existing research has proposed numerous feature selection algorithms and knowledge-guided learning approaches. However, there is a distinct research gap: there is no generic, selector-independent framework that can systematically incorporate clinical domain knowledge as a post-selection refinement step across multiple feature selection techniques.

This is where the Decoupled Operator-Driven Architecture (DODA) introduces a paradigm shift.

Core Paradigm Shift & Defensible Claim. Our research position is explicitly distinct: statistical feature selection is not fundamentally wrong. Instead, we claim that statistical feature selection optimizes statistical relevance, whereas clinical decision support requires both statistical and domain relevance.

3. Research Aim, Objectives, & Key Questions

Research Aim. To develop and evaluate a modular domain-aware framework — built upon a Decoupled Operator-Driven Architecture — that refines statistically selected features using clinical knowledge in order to improve the clinical relevance and trustworthiness of machine learning models.

Research Objectives (RO)

1. **RO1:** Evaluate multiple statistical feature selection methods on several clinical datasets.
2. **RO2:** Quantify how often statistically selected features differ from clinically important features.
3. **RO3:** Develop a domain-aware refinement layer (DODA) that adjusts feature rankings using clinical knowledge.
4. **RO4:** Compare baseline feature selection with DODA-enhanced feature selection across multiple datasets.
5. **RO5:** Evaluate whether clinically guided feature refinement improves clinical relevance, maintains predictive performance, increases feature stability, and improves model interpretability.

Research Questions (RQ) & Hypotheses

- **RQ1:** Do conventional feature selection algorithms consistently identify clinically important variables?
- **RQ2:** Can domain knowledge recover clinically important features overlooked by statistical feature selection?
- **RQ3:** Does domain-guided feature refinement improve model interpretability without sacrificing predictive performance?
- **RQ4:** Can a selector-independent framework generalize across multiple datasets and feature selection methods?

Null Hypothesis (H_0): There is no significant difference between statistical feature selection and clinically guided feature selection.

Alternative Hypothesis (H_1): Clinically guided feature refinement produces feature subsets that are more clinically meaningful while maintaining comparable predictive performance.

4. Methodology & Evaluation Pipeline

The framework operates via a unified multi-stage experimental architecture, validating the selector-independent design across varied medical domains.

DODA Architectural Pipeline

Dataset → Feature Selection → Statistical Ranking

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Clinical Knowledge Layer (DODA) → Adjusted Ranking → Top-K Features
→ ML Model

Step-by-Step Implementation

1. **Step 1 (Select Datasets):** Select four public clinical datasets (e.g., Diabetes, Heart Disease, Breast Cancer, and Stroke / CKD).
2. **Step 2 (Conventional Selection):** Run conventional feature selection algorithms: ANOVA, Mutual Information, LASSO, mRMR, Random Forest Importance, and Boruta.
3. **Step 3 (Baseline Models):** Train baseline models (Logistic Regression, Random Forest, XGBoost) and measure performance across Accuracy, Precision, Recall, F1, and AUC.
4. **Step 4 (Apply DODA):** Apply the Decoupled Operator-Driven Architecture layer to adjust statistical rankings based on clinical knowledge.
5. **Step 5 (Pipeline Replication):** Train the exact same models again over the identical pipeline, varying only the updated feature rankings.
6. **Step 6 (Comprehensive Comparison):** Evaluate and cross-examine the results along predictive, clinical, explainability, and stability axes.

5. Multi-Dimensional Validation Framework

To challenge the typical limitations of medical AI literature, which frequently isolates evaluation strictly to predictive metrics, this proposal evaluates each feature subset across four complementary axes:

Dimension	Target Evaluation Metrics & Verification Strategy		
		Baseline FS	DODA
Predictive Performance	Accuracy, Precision, Recall, F1, and ROC-AUC.	✓	✓
Clinical Relevance	Clinical relevance score, number of clinically important biomarkers retained, and agreement with expert knowledge.	—	✓

Dimension	Target Evaluation Metrics & Verification Strategy	Baseline FS	DODA
Feature Stability	Robustness tracking across multiple folds via Jaccard similarity and rank correlation.	—	✓
Explainability	SHAP comparison and analysis of feature ranking changes.	—	✓

6. Expected Contributions

Unlike existing approaches that embed domain knowledge into feature engineering or model training, this work introduces a selector-independent post-selection refinement framework. By leveraging a Decoupled Operator-Driven Architecture, this method is capable of enhancing any statistical feature selection technique through modular domain knowledge integration. This provides a robust strategy for medical AI deployments, allowing frameworks to retain high accuracy while vastly improving human-expert alignment and trust.